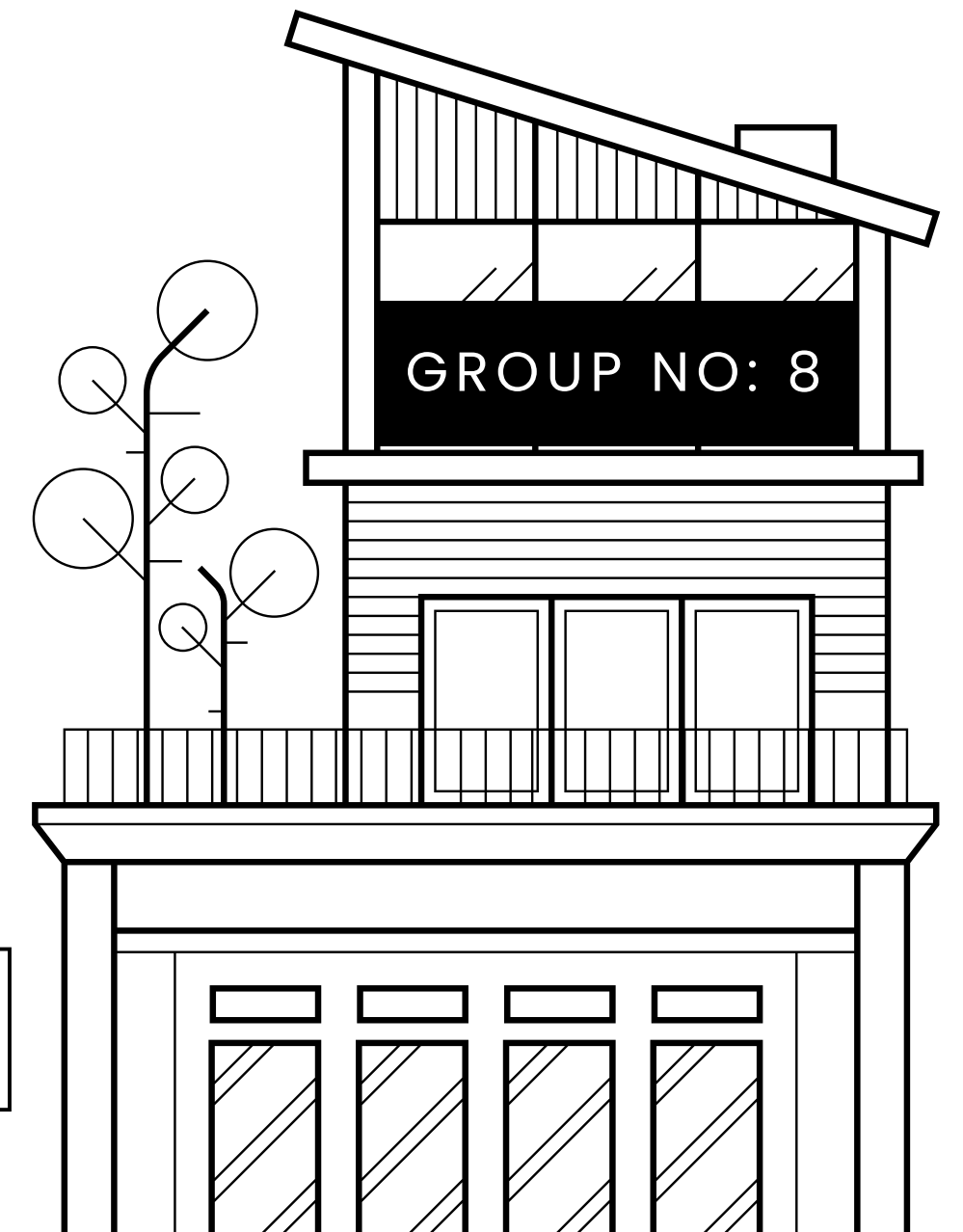


GenArchAI: An AI-Powered Platform for Generative Architectural Design

“Transforming architectural design through intelligent automation”

Presented by: Ameen Ali [MES22AD011]
Farah Fahmi [MES22AD019]
Fathima Sana [MES22AD024]
Gayathri Krishna [MES22AD026]

Guided by: Mr. Sherikh K K
Assistant Professor,
ADS Department



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INTRODUCTION

GenArchAI is an AI-powered platform that makes architectural design smarter for new buildings and renovations. It creates floor plans, 3D visuals, and style options automatically, saving time and effort.

Key Highlights:

- Design Generation: Upload land sketches AI makes new plans, elevations, and exteriors.
- Renovation & Style: Compare old vs. new designs and pick from different styles.
- Smart Budget: Get instant cost estimates and adjust designs to fit budgets.
- AI Chatbot: Guides users with design tips, budget help, and renovation advice.

OBJECTIVES

- Automate time-consuming architectural design tasks
- Generate high-quality design alternatives quickly
- Provide real-time cost estimation and budget optimization
- Improve efficiency in resource planning and decision-making
- Make advanced design tools accessible to all users
- Enable interactive AI-based design assistance

PROBLEM STATEMENT

- Architectural design process is manual, slow, and time-consuming
- Clients face difficulty in visualizing multiple design options
- Cost estimation is often delayed and inaccurate, leading to budget overruns
- Limited AI support for quick renovation and redesign
- Communication between architects and clients is time-intensive
- Lack of interactive tools for real-time design assistance

LITERATURE SURVEY (CONT...)

Aspect	Insights from Literature	Planned Implementation in Project
Generative Design	GANs and sketch-based AI create layouts and visuals.	Use FLUX.1 AI model for automated floor plans, elevations, and exterior design generation.
3D & Renovation	Parametric + BIM enable flexible modeling and visualization.	Apply Gemini Vision + AI image generation for renovation and style transformation.
Cost Estimation	ANN shows high accuracy in predicting cost and duration.	Implement a smart cost estimation engine using real-world construction parameters and datasets.
AI Chatbot	Chatbots help refine ideas and guide design choices.	Use LLaMA 3.1-powered chatbot for design guidance, budgeting, and renovation support.
All-in-One System	Previous works focused on individual aspects.	Integrate design generation, renovation, cost estimation, and chatbot assistance into a single platform.

PROPOSED SYSTEM

GenArchAI uses generative AI, smart APIs, and custom datasets to make architectural design easier, faster, and cost-effective.

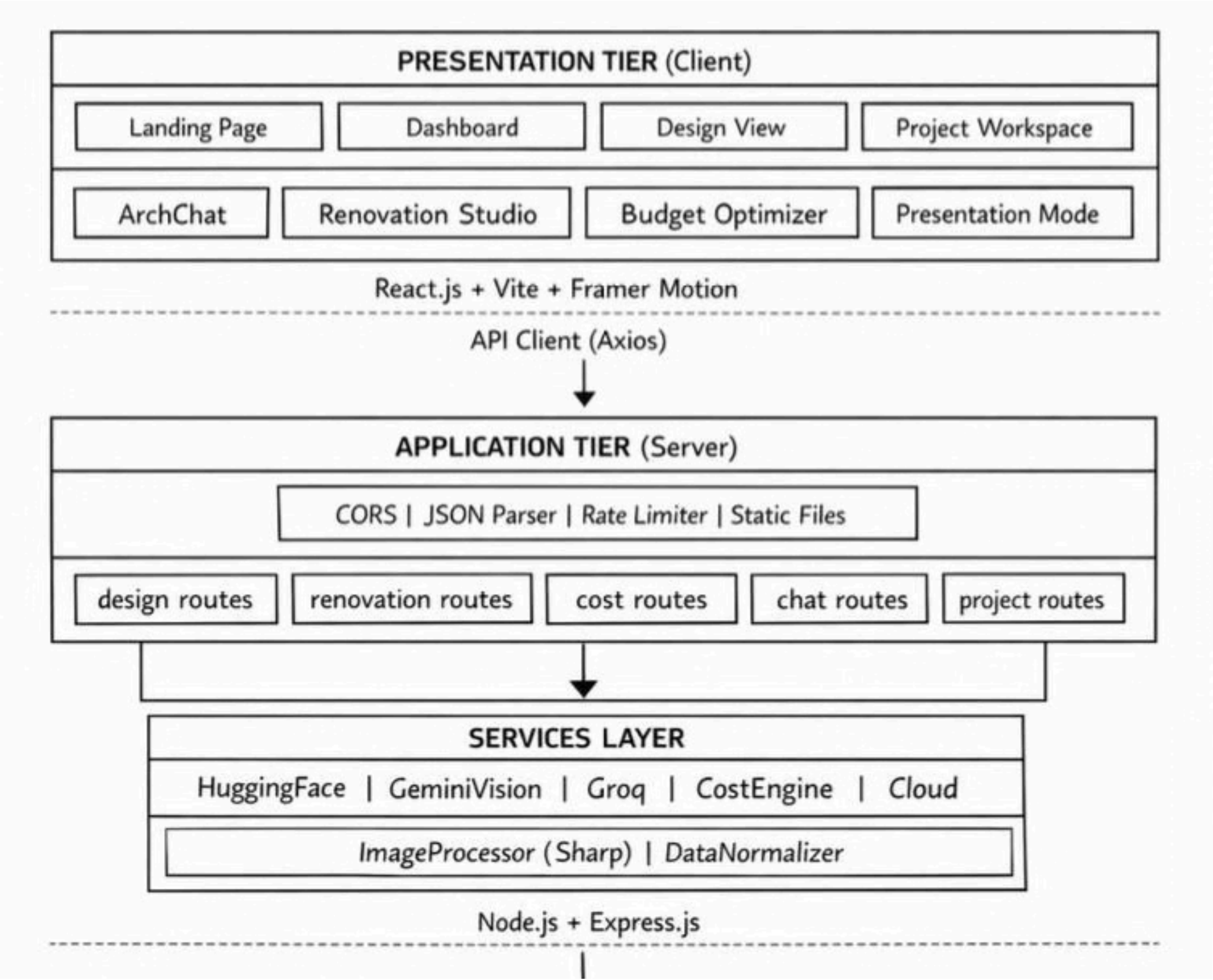
Core components:

- **AI Design Generation** – Generates architectural designs
- **Sketch Analysis** – Converts sketches into structured designs
- **Renovation Module** – Creates redesigned layouts
- **Cost Estimation** – Calculates construction cost
- **AI Chatbot** – Provides design and budget guidance
- **Data & Storage** – Manages data and generated images

METHODOLOGY

- Designed a three-tier client–server architecture
- Divided system into modules: Design, Renovation, Cost, Chatbot, Optimization
- Integrated AI services for image generation, vision analysis, and NLP
- Developed using React.js (frontend) and Node.js & Express (backend)
- Used Firebase Firestore and cloud storage for data management
- Performed module testing, integration, and deployment

SYSTEM ARCHITECTURE



SYSTEM ARCHITECTURE (CONT...)

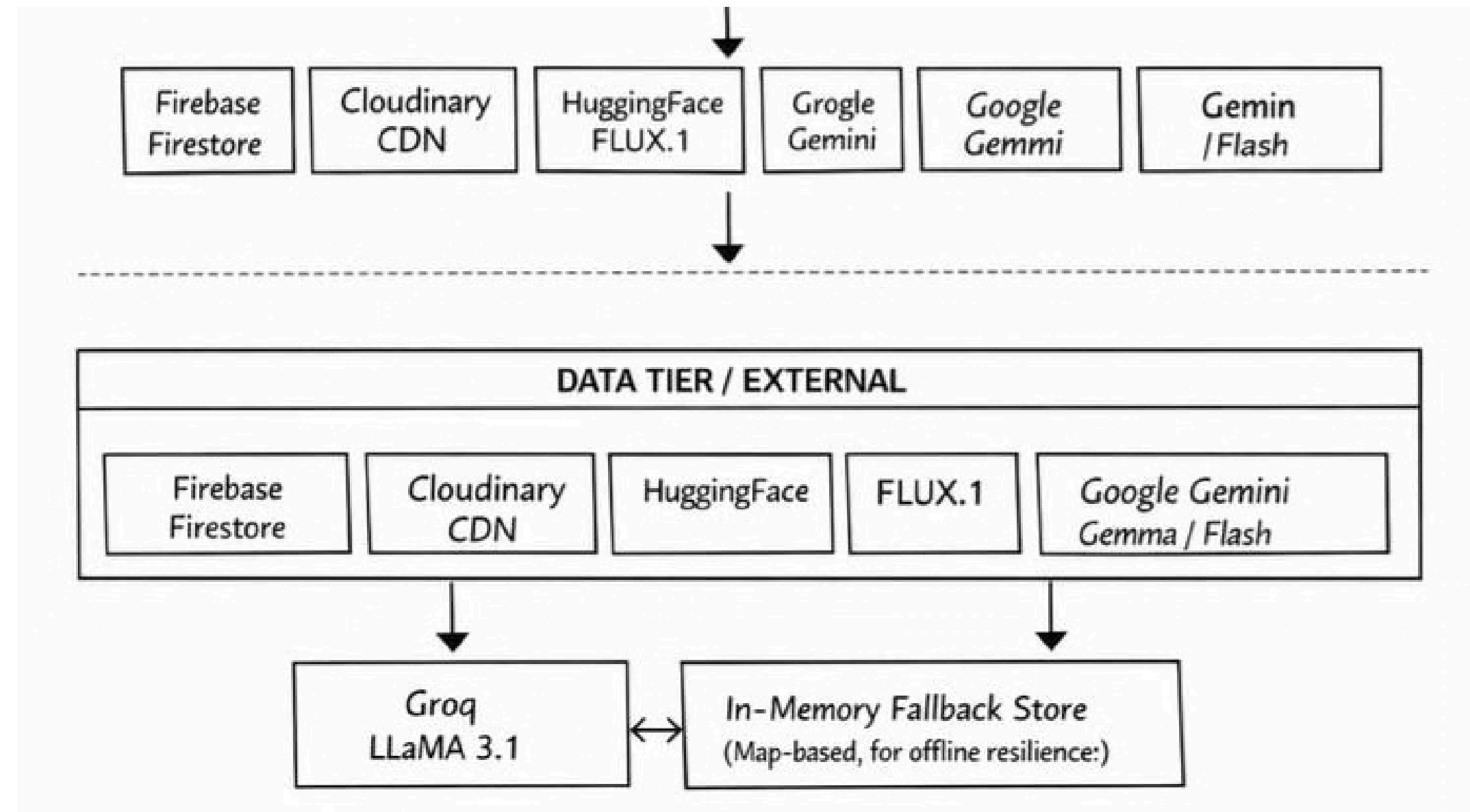


fig 1 : System Architecture Flow Diagram

WORKING PRINCIPLE

1. User Input & Project Setup

- User opens the web application
- User creates or selects a project
- User enters a text prompt or uploads a sketch

2. Backend Request Processing

- Request is sent to the backend server
- Vision API analyzes the sketch (if uploaded)
- Prompt is sent to the image generation model

WORKING PRINCIPLE (CONT...)

3. AI Design Generation & Storage

- AI generates the architectural design
- Generated image is stored in cloud storage
- Design details are stored in the database

4. Cost Estimation & Budget Optimization

- Cost engine calculates construction cost
- Budget optimizer suggests cost changes
- System estimates material and labor requirements

WORKING PRINCIPLE (CONT...)

5. Design Assistance & Renovation

- Renovation module generates new style designs
- Chatbot provides design guidance
- AI suggests alternative design layouts based on user preferences

6. Result Display & Data Management

- Frontend displays design and cost
- User can compare designs
- User can export PDF report

TOOLS AND APPROACHES

Frontend

- React.js – User interface and interaction

Backend

- Node.js & Express – API handling and system logic

AI Technologies

- FLUX.1 – Architectural design generation
- Gemini Vision – Sketch analysis
- LLaMA 3.1 – Chatbot and assistance

TOOLS AND APPROACHES (CONT..)

Database & Storage

- Firebase Firestore – Data storage
- Cloudinary – Image storage

Integration

- Socket.IO – Real-time communication
- PDFKit – Report generation

DATASET

Cost Estimation Dataset

- Construction cost reference data
- Includes parameters such as area, floors, quality, and location
- Material & labor cost parameters
- Includes cost components like foundation, structure, electrical, plumbing, and finishing
- Used in Cost Engine for cost calculation
- Used for budget comparison and optimization

USER INTERFACE

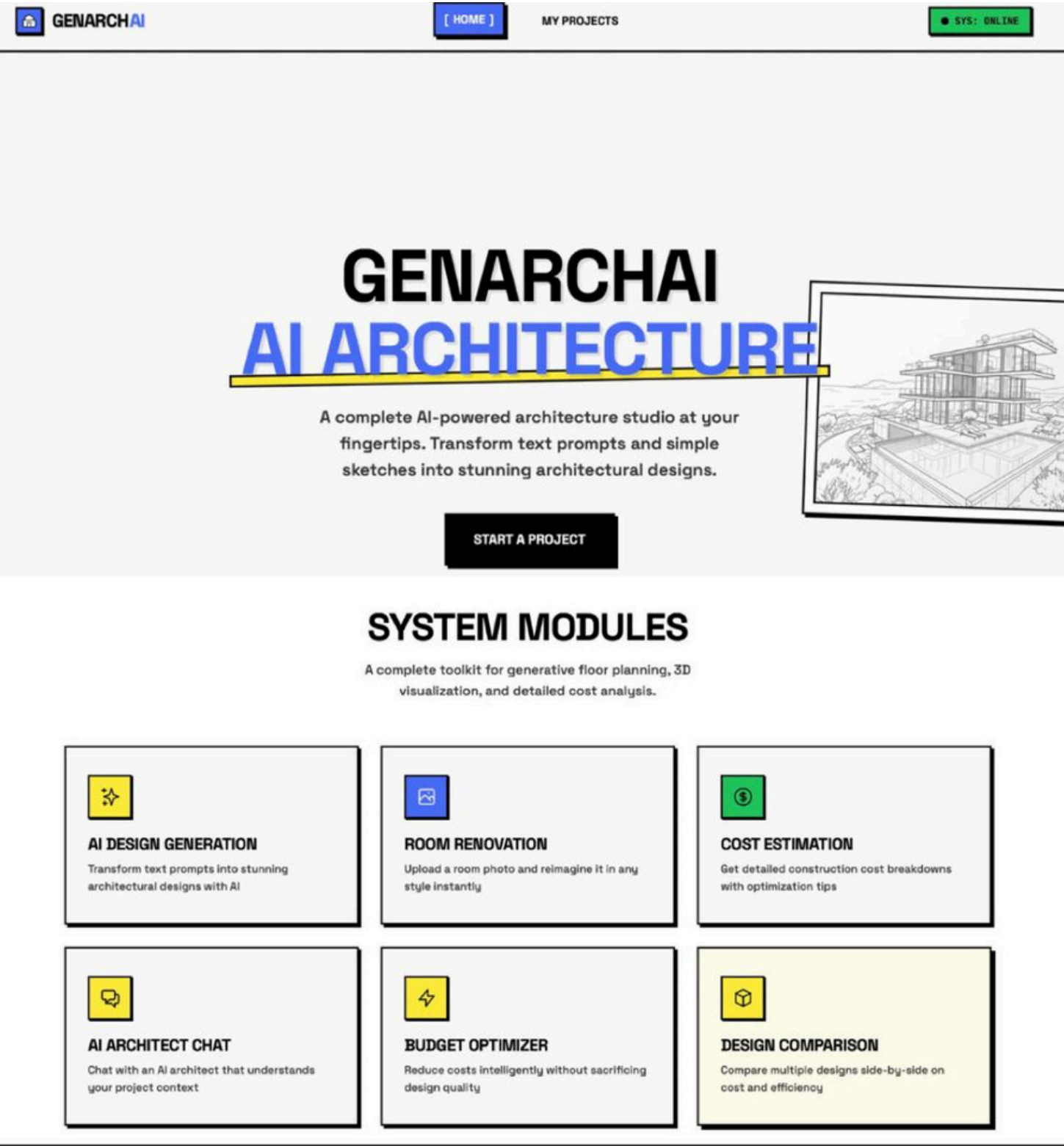


fig 2 : Landing page and System Modules

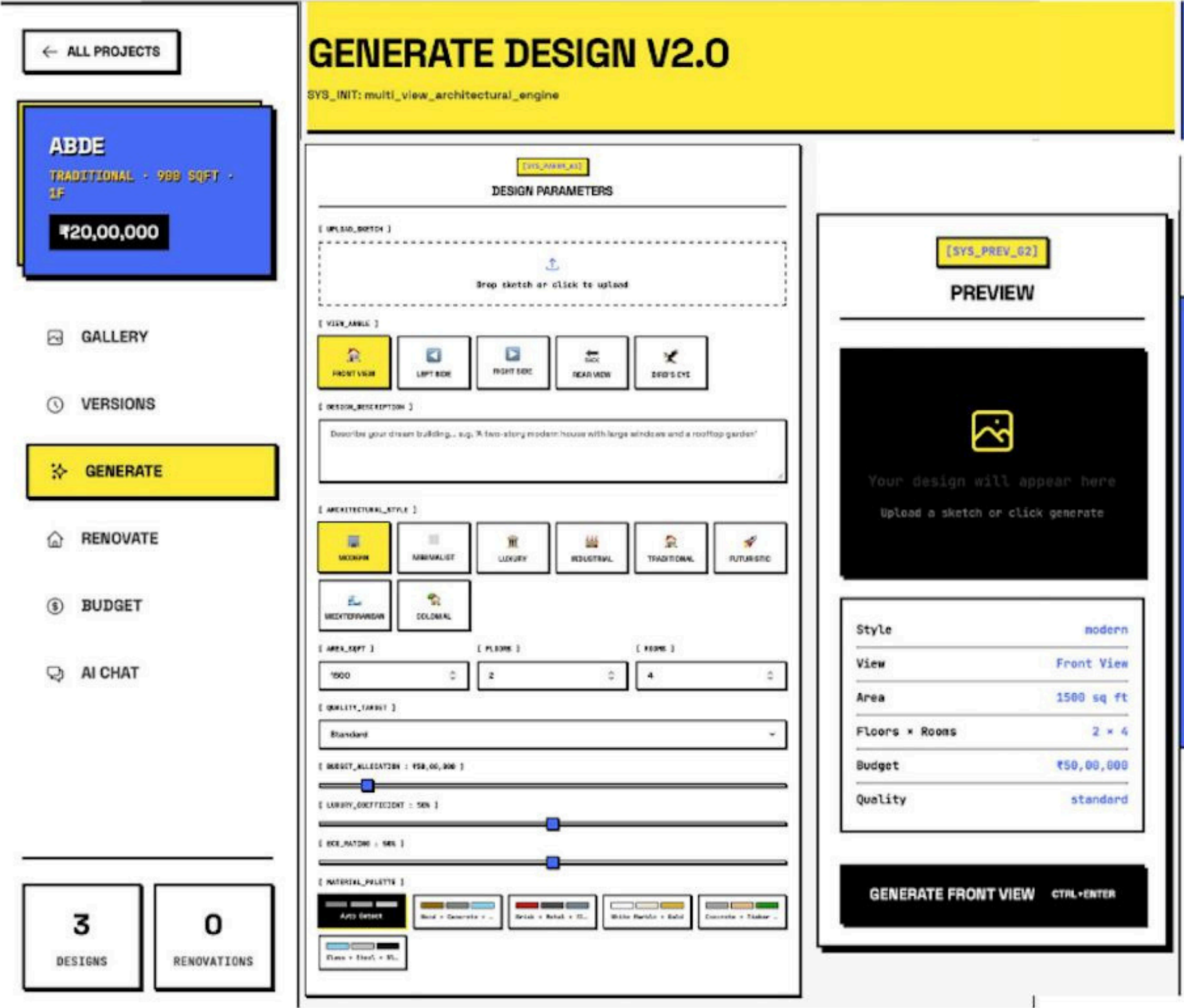


fig 3 : AI Design Generation Page

USER INTERFACE (CONT..)

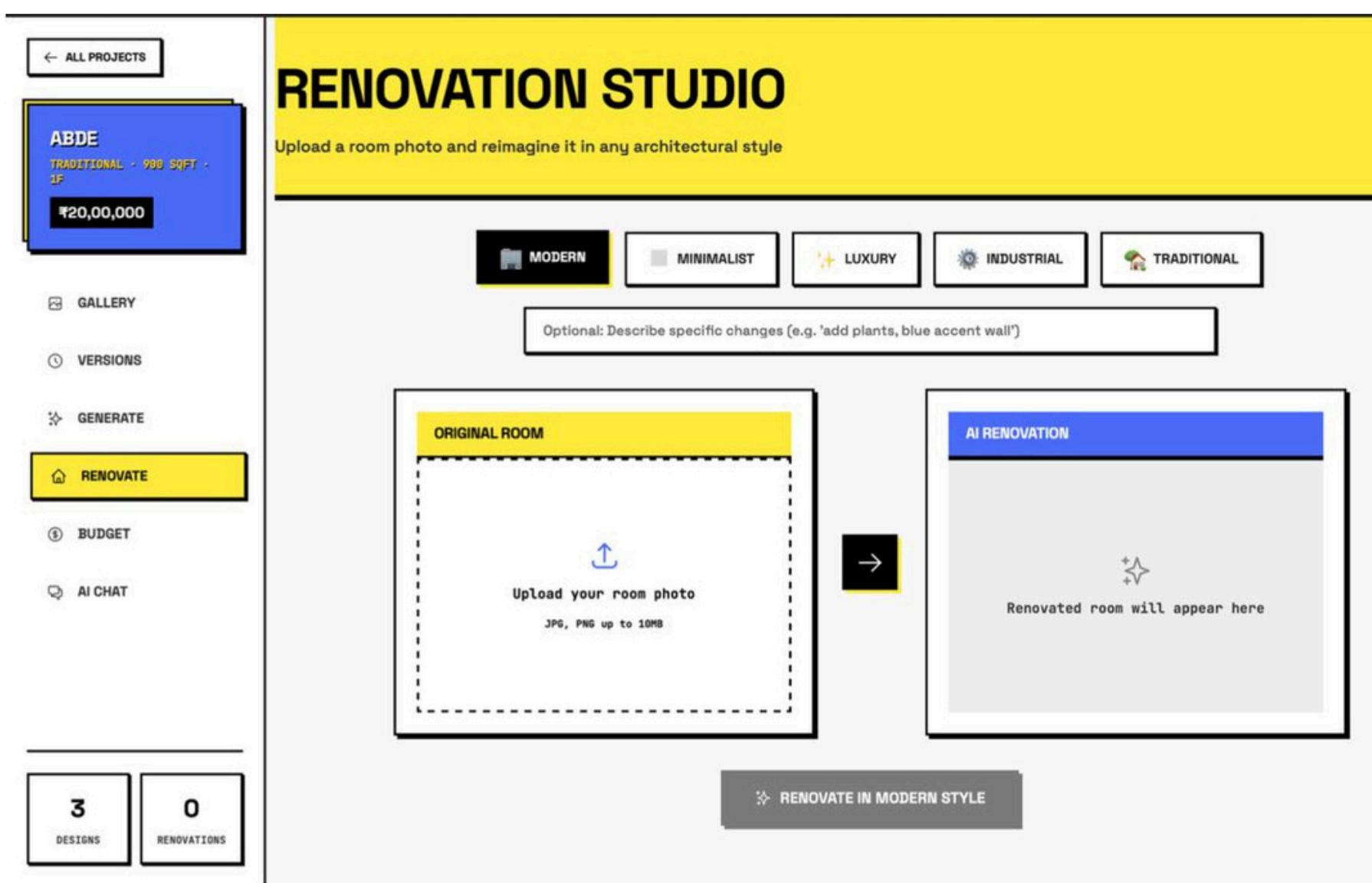


fig 4: Renovation Studio Interface

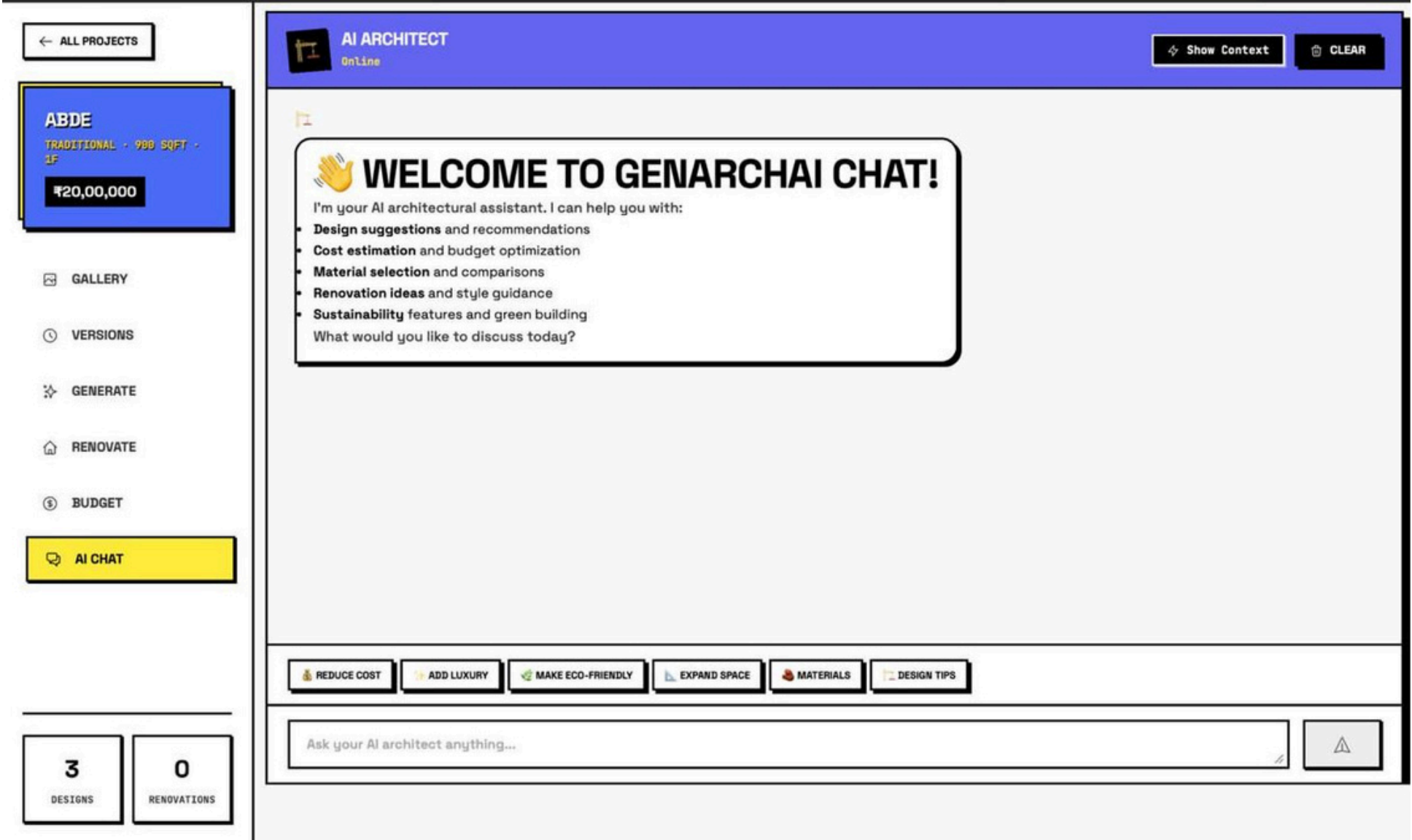


fig 5 : AI Architect Chat Interface

USER INTERFACE (CONT..)

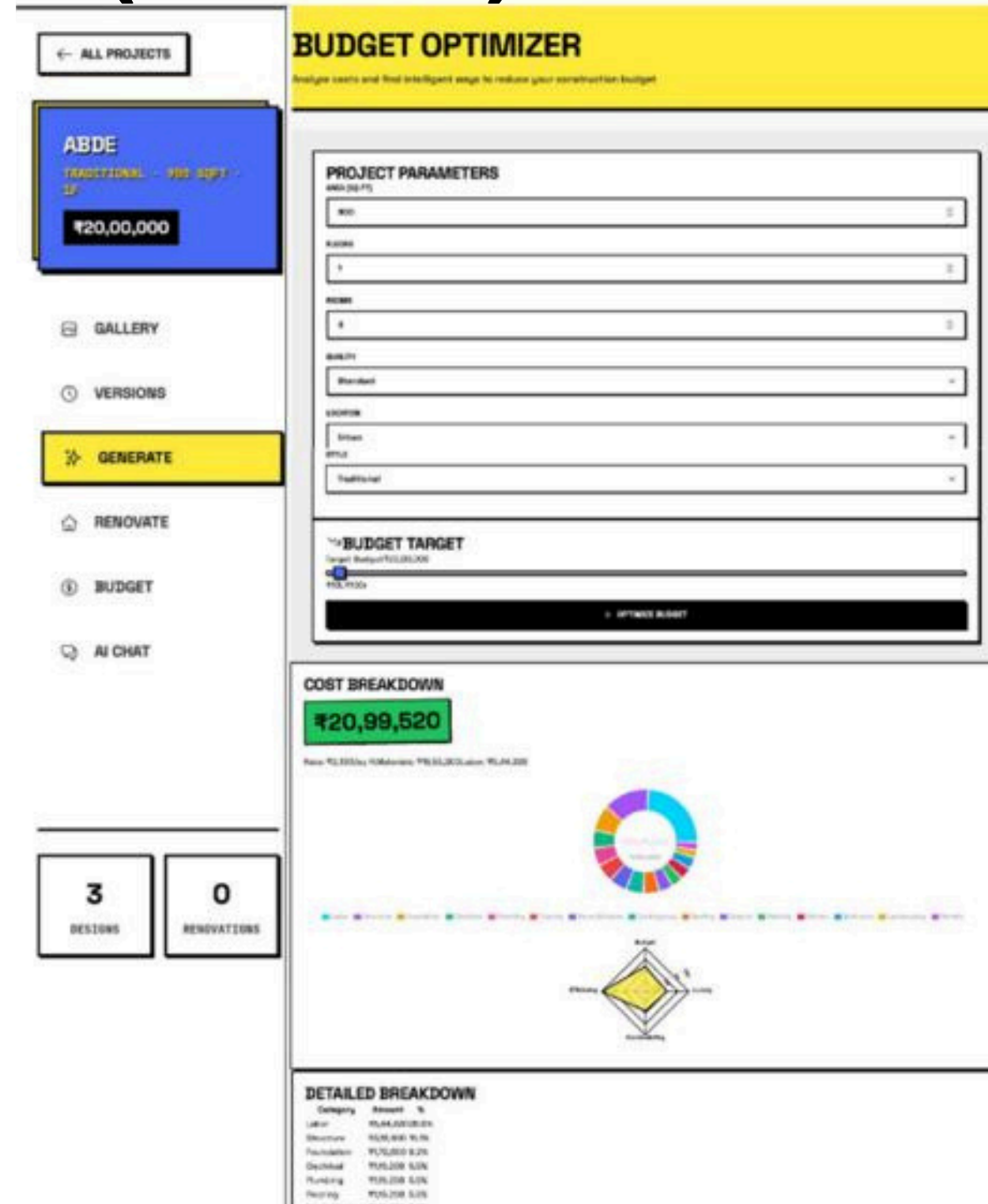
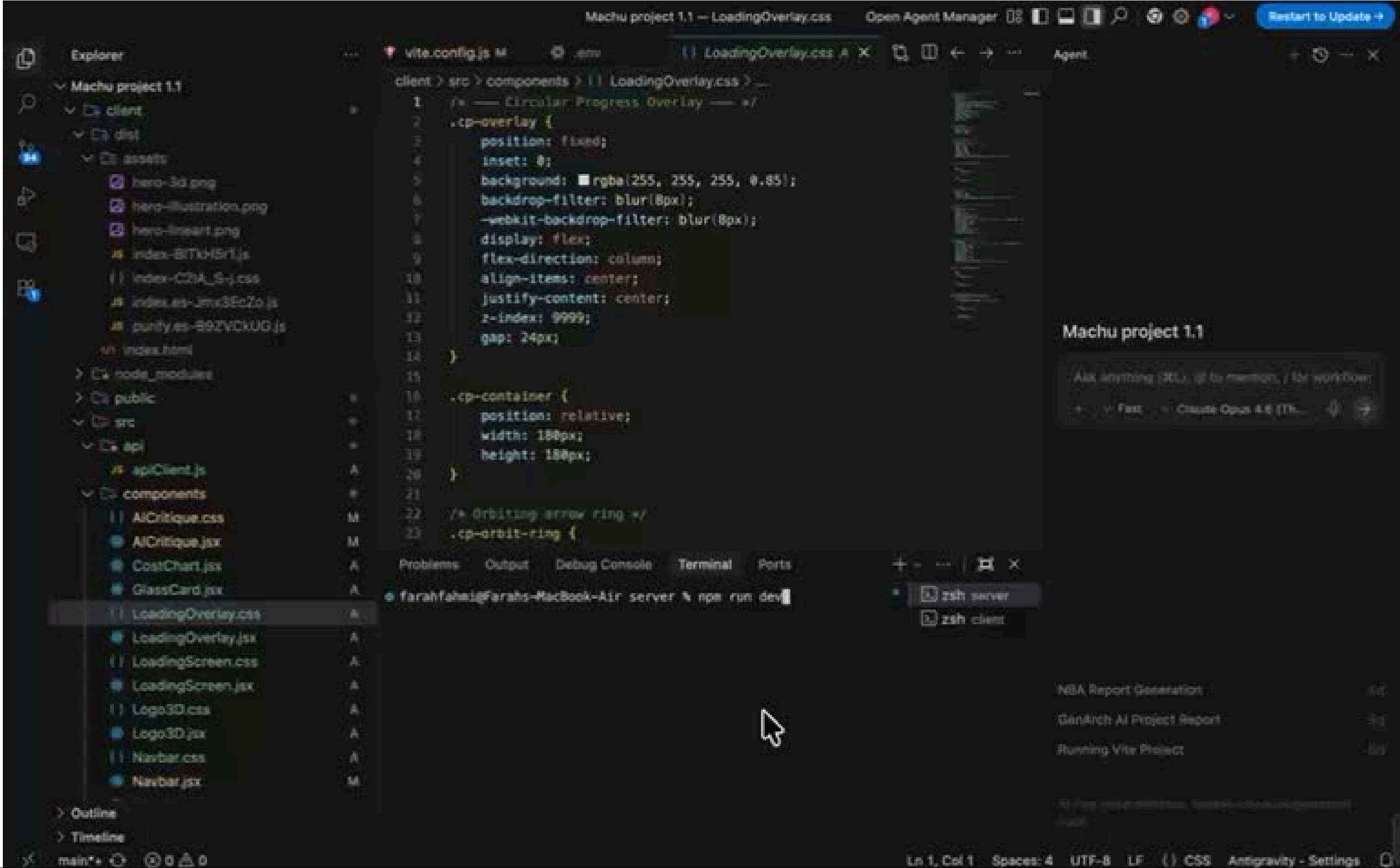


fig 6: Budget Optimizer

WORKING DEMO



RESULT AND ANALYSIS

System Performance Metrics

Parameter	Value	Observation
Design Generation Time	~5–12 sec	Fast AI-based generation
Sketch Processing Time	~3–8 sec	Efficient vision analysis
Cost Estimation Time	< 1 sec	Instant calculation
API Response Time	~2–5 sec	Stable performance
AI Chatbot	~1–3 sec	Fast conversational response
Overall System Response	~8–15 sec	Real-time interaction

RESULT AND ANALYSIS (CONT..)

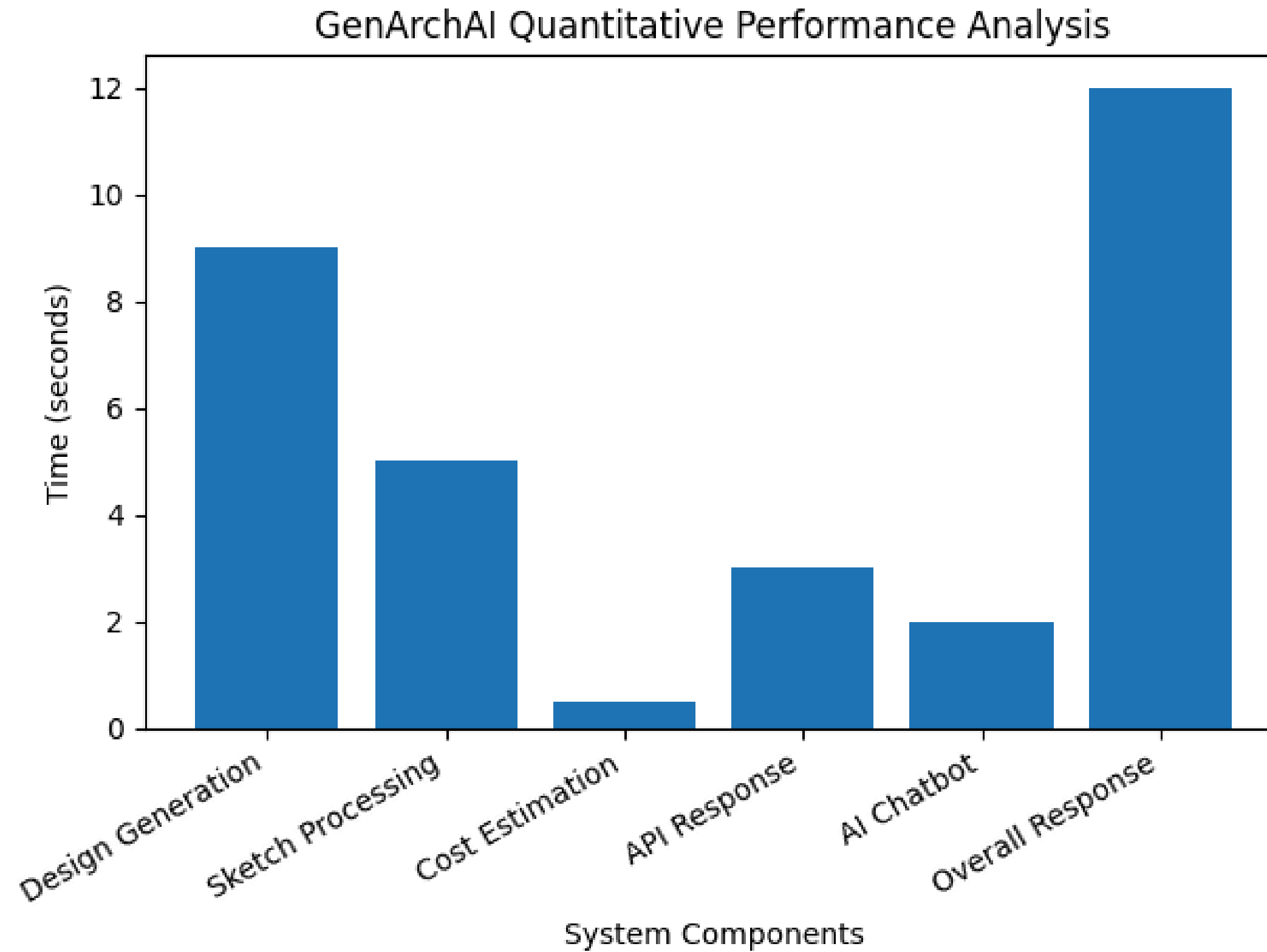


fig.7: Quantitative analysis graph

RESULT AND ANALYSIS (CONT..)

Accuracy & Effectiveness

- Cost estimation accuracy: ~80–90% (approximate real-world alignment)
- Design output quality: High (multi-view consistency achieved)
- Chatbot response relevance: Context-aware and interactive

User Feedback (Testing Phase)

- Users found the system easy to use and interactive
- Improved design visualization and decision-making
- Positive feedback on speed and accessibility

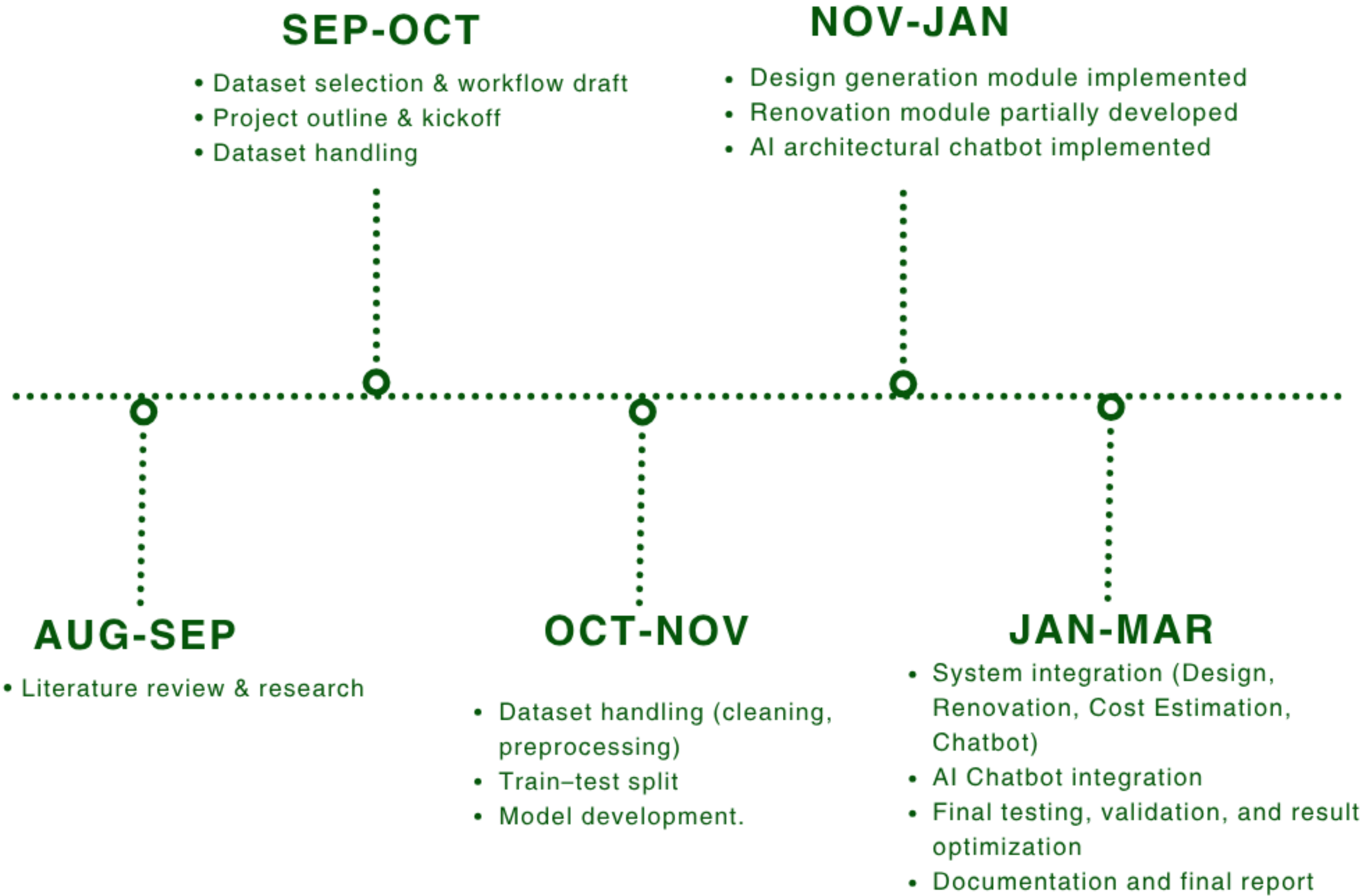
IMPACTS & ADVANTAGES

- Makes architectural design accessible to non-experts
- Reduces design time from weeks to seconds
- Provides real-time cost estimation
- Improves visualization and decision-making
- Supports homeowners, students, and developers
- Acts as a cost-effective alternative to traditional methods

LIMITATIONS

- AI-generated designs may require expert validation
- Depends on external APIs and internet connectivity
- Cost estimation is approximate, not exact

TIME LINE



TEAM MEMBERS CONTRIBUTION

Ameen Ali (Backend & Cost Estimation).

- Node.js backend, Cost Engine, Firestore integration

Farah Fahmi (Design Generation).

- FLUX.1 integration, Gemini Vision, design APIs

Fathima Sana – Frontend & Chatbot

- React UI, API integration, LLaMA chatbot

Gayathri Krishna O S – Renovation & Testing.

- Style module, testing, validation

All Members

- Integration, testing, documentation, report

FUTURE SCOPE

- Improve AI design generation quality and realism
- Add advanced 3D visualization features
- Enhance cost estimation accuracy using larger datasets
- Extend AI chatbot capabilities for better guidance
- Support more building types and project categories
- Improve system scalability and cloud deployment

CONCLUSION

- GenArch integrates AI with architecture to simplify design and planning.
- It provides intelligent suggestions for layouts, costs, and project decisions.
- The system reduces manual effort and saves time for users.
- A user-friendly interface makes it accessible for beginners
- It combines multiple features like design, estimation, and management in one platform.

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Thank You

Any Queries?

